Curriculum Vitae

Emanuele Bernardi

Applied mathematician working on phenotype-structured integro-differential models, combining analytical methods and numerical simulations to study evolutionary and epidemiological dynamics.

Date and place of birth: 19 June 1997, Pistoia (Italy)
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Research Experience

- **PhD in Pure and Applied Mathematics**, Politecnico di Torino – Università di Torino (23 February 2026), *cum laude*.
  Thesis: *Phenotype-structured integro-differential models: derivation and qualitative analysis*. Supervisors: Prof. Tommaso Lorenzi, Prof. Andrea Tosin.
- **Research Assistant**, University of Pavia, Department of Mathematics "F. Casorati" (2021–2022).

Education

- **Master in Science and Technology**, IUSS Pavia (28 November 2022), grade 100/100 *cum laude*.
  Thesis: *Numerical and statistical analysis for hyperspectral imaging improvement*. Supervisors: Prof. Andrea Taramelli, Dr. Margherita Righini.
- **Master's Degree in Mathematics**, University of Pavia (20 July 2021), grade 110/110 *cum laude*.
  Thesis: *Kinetic model for wealth distribution in presence of epidemic dynamics with asymptomatic cases*. Supervisors: Prof. Mattia Zanella, Prof. José A. Carrillo.
- **Bachelor's Degree in Mathematics**, University of Pavia (19 September 2019), grade 110/110 *cum laude*.
  Thesis: *Study of a Non-linear Evolution Problem*. Supervisor: Prof. Giulio F. Schimperna.

Visiting Experience

- February-March 2025: Visiting PhD student, Université Paris Cité, IMJ-PRG, Paris.

Publications

1. E. Bernardi, T. Lorenzi, M. Sensi, A. Tosin. *Heterogeneously Structured Compartmental Models of Epidemiological Systems: From Individual-Level Processes to Population-Scale Dynamics*. *Studies in Applied Mathematics*, 155(2): e70091, 2025. DOI: https://doi.org/10.1111/sapm.70091
2. E. Bernardi, T. Lorenzi, A. Tosin. *Derivation and quasi-invariant asymptotics of phenotype-structured integro-differential models*. *arXiv preprint arXiv:2510.15646*, 2025. https://arxiv.org/abs/2510.15646
3. E. Bernardi, L. Pareschi, G. Toscani, M. Zanella. *Effects of vaccination efficacy on wealth distribution in kinetic epidemic models*. *Entropy*, 22(2):216, 2022. DOI: https://doi.org/10.3390/e24020216

Grants and Awards

- Research scholarship "Machine Learning for Industry 4.0 and Digital Healthcare", Camelot Biomedical Systems (Dec 2021 - Nov 2022).
- Scholarship for the Master's Degree in Mathematics, University of Pavia (A.Y. 2019-2020).
- Exchange Scholarship, Collegio Ghislieri - St. Hugh's College, Oxford (Spring 2021).

Teaching Experience

Teaching Assistant in Undergraduate Courses

- Calculus for Engineering, BSc, Politecnico di Torino (2024).
- Calculus for Industrial Engineering, BSc, University of Pavia (2022).
- Mathematics for Geology, BSc, University of Pavia (2020).

Teaching Assistant in College Courses

- "Geometry 1", Collegio A. Volta, preparation of a complete workbook with solutions (2019).

Self-contained Mini Courses

- "Introduzione all'algebra lineare ed applicazioni", Collegio Ghislieri (2020).

Conferences, Workshops & Schools

- June 2024: Summer School **"From Cells to Tissues: Models, Analysis and Applications"**, contributed talk: *Derivation and quasi-invariant asymptotics of phenotype-structured integro-differential models*, Como, Italy.
- June 2024: **12th Summer School on Methods & Models of Kinetic Theory**, poster: *Derivation and quasi-invariant asymptotics of phenotype-structured integro-differential models*, Pesaro, Italy.
- January 2024: Workshop **"Integrated Mathematical Approaches to Socio-Epidemiological Dynamics"**, contributed talk: *Derivation and asymptotic analysis of integro-differential models of phenotypic evolution*, Trento, Italy.
- December 2022: Workshop **"From Kinetic Theory to Data Science and Related Topics"**, Pavia, Italy.
- September 2022: **XLVII Summer School on Mathematical Physics**, contributed talk: *Effects of vaccination efficacy on wealth distribution in kinetic epidemic models*, Ravello, Italy.
- June 2022: **11th Summer School on Methods & Models of Kinetic Theory**, poster: *Effects of vaccination efficacy on wealth distribution in kinetic epidemic models*, Pesaro, Italy.
- September 2021: **1st Young Applied Mathematicians Conference**, contributed talk: *Kinetic models for wealth distribution in presence of epidemic dynamics with asymptomatic cases*, Santa Maria di Leuca, Italy.

Languages and Computer Skills

- **Languages:**
  Italian (mother tongue), English (C1, IELTS 7.5), French (basic knowledge).
- **Software:**
  Excellent knowledge of MATLAB; basic knowledge of RStudio and Python.

Academic Service and Outreach

- Founder and President (2018-2021), Student Association **Matematica Pavese**, University of Pavia. Organised seminars and meetings connecting students with academic and professional speakers.

- Member and tutor (2018-2022), **Accademia dei Giovani per la Scienza**, promoting scientific engagement among high-school students.
- Member, **Ghislieri Scienza** (2016-2018), organiser of the scientific festival *Indiscienza*.